

Causal Erasure: the Process-Rate Conditional Coordinate

GO-14 statement v0.5 — Theorems R, C and T (the convex program, and the n -transfer lemma), certified two-sided anchors and a certified within-class ladder, under the U-independence hypothesis and the boundary-charge hypothesis (H^*)

Geometric Observation program

v0.5 — August 2026 (v0.4 superseded: Theorem T added — exact subadditivity, Fekete existence of the limit and the *unconditional* lower transfer side — together with the numbered hard hypothesis (H^*) carrying the boundary-charged reverse direction, the conditional $\Delta = 0$ plateau corollary, and the executed within-class certification of the diagonal ladder; six R-IND-5 transfer restatements in force, including the refutation of “ $\leq \kappa$ per side” as an $\mathcal{F}_0$ lemma. v0.4 history: Theorems R and C with the certified anchor table and the sandwich-margin interval, seven R-IND-5 restatements. v0.2 history: process-limit conjecture, hard hypothesis, dated erratum to Theorem 1(v); the v0.1 sign error was corrected at v0.2)

Abstract

GO-12 settled the single-record causal-prefix eraser (the Kalman family); the process-rate object — block records erased cell by cell, each against only the causal prefix of an aging reference — was promoted to a first theorem at v0.2 (netted, GO-P-2026-076). v0.3 added the process-limit face: the $n \rightarrow \infty$ limit of the coordinate exists at $O(1/n)$, closes at the smoothing-pole rate $\lambda_s^{2\Delta}$ with a constant that is *not* matched by any tested causal-conditioning spectrum, and is driven by an explicit, testable record mechanism (horizon-matched V -cancellation). This revision adds the lower-bound theory: in moment coordinates (H, Γ) the family optimization is a jointly convex program — Theorem R (a schedule-general moment-coordinate representation of nL_a) and Theorem C (joint convexity, so every KKT point is the global family minimum and strong duality yields computable lower bounds). All seven full-space anchors are now *certified two-sided* (widths 3.45×10^{-6} – 1.7×10^{-5}), and the sandwich margin is a certified interval: $\min L_a(0) - \text{block}_{16} \in [0.0317872, 0.0317906]$. This revision adds the n -transfer theory. Theorem T is unconditional: block concatenation gives exact subadditivity $f(n_1 + n_2) \leq f(n_1) + f(n_2)$, so by Fekete the process limit $L^\infty = \lim_n \phi_n = \inf_n \phi_n$ *exists*, and the lower transfer side $0 \leq \phi_n - L^\infty$ is a theorem — every certified finite- n value bounds the limit from above. The *reverse* direction is not: the boundary charge is carried by a numbered hard hypothesis (H^*) on dyadic-chain optimizers, because “ $\leq \kappa$ per side” is *false* as a family statement (D -feasible U -independent counterexamples reach 26κ ; no universal constant exists). Under (H^*) the $\Delta = 0$ plateau over the causal-spectral allocation becomes a 3.65×10^{-3} -margin statement about the limit itself rather than about the $O(1/n)$ model. The diagonal ladder is now certified *within class*. Two scoping results are load-bearing and stated as hard terms: (i) the record family — jointly Gaussian with (V, Y) , *independent of the reference noise U* — is a numbered hypothesis on the chain rule and on every minimum quoted here (U-coupled records break the identity and invert the sandwich); (ii) all numerical minima remain *class-conditional or finite- n upper bounds*; nothing here relies on them being global.

1 Setting

Sources as in GO-12: V AR(1) with pole a , unit stationary variance; $Y_t = \rho V_t + N_t$ with $\text{Var}(Y) = 1$; reference $S_t = V_t + U_t$, $\text{Var}(U) = \tau^2$. A block record $\hat{Y}^n$ of Y^n (per-symbol distortion $\frac{1}{n} \sum_t E(Y_t - \hat{Y}_t)^2 \leq D$) is erased cell by cell; the eraser of cell t holds $S^{t+\Delta}$ and the already-erased cells $\hat{Y}^{t-1}$. The coordinate is

$$L_a(\Delta) = \frac{1}{n} \sum_{t=1}^n I(T^n; \hat{Y}_t \mid S^{t+\Delta}, \hat{Y}^{t-1}), \quad T = (V, Y).$$

Hypothesis 1 (U-independence — hard, load-bearing). *The record family $\mathcal{F}_0$: $\hat{Y}^n$ jointly Gaussian with (V^n, Y^n) and independent of U^n given (V^n, Y^n) ; concretely $\hat{Y} = A_y Y + A_v V + Z$ with $Z \perp (V, N, U)$ (no U -coupling, $A_u = 0$). Theorem 1 (the chain rule and every item), Conjecture 1, and every minimum quoted in this document are stated under this hypothesis; no seal of the process-limit face is valid without it.*

Extension and counter-values (R-IND-5, on record). The chain rule of Theorem 1(i) extends *exactly* to U -coupled records $\hat{Y} = A_y Y + A_v V + A_u U + Z$ if and only if the U -coupling is Δ -lag causal, $A_u[t, s] = 0$ for $s > t + \Delta$. Outside that: the identity is *false* (measured residuals 0.015–0.40 at random U -couplings; 4.59 bits at an optimizer endpoint), and the minima collapse — the trivial Δ -causal U -record $\hat{Y}_t = 0.54 Y_t + 0.23 S_t + \mathcal{N}(0, 0.1625)$ is D -feasible and achieves $L_a(0) = 0.539536 < 0.566758$ (the family min), and a general U -coupled certified-feasible record achieves $L_a(0) = 0.092864 < \text{block}_{16} = 0.534968$: the sandwich of item (iii) is *inverted*. The U -coupled variant is a different, far cheaper coordinate and is *open*.

2 The theorem

Write $\lambda_s = a(1 - K)$ for the steady-state Kalman *smoothing* pole (K the steady Kalman gain of S on V); at the netted parameters $(a, \rho^2, \tau^2) = (0.8, 0.49, 0.4)$, $\lambda_s = 0.354554$, $\lambda_s^{-2} = 7.955$.

Theorem 1 (Causal erasure, first netting — GO-P-2026-076). *For every jointly Gaussian record on the window $t = 1, \dots, n$ satisfying Hypothesis 1:*

(i) **Exact chain rule.** $n L_a(\Delta) = I(T^n; \hat{Y}^n \mid S^n) + C_\Delta$ with

$$C_\Delta = \sum_t I(\hat{Y}_t; S_{>t+\Delta} \mid S^{t+\Delta}, \hat{Y}^{t-1}) \geq 0$$

the smoothing-leakage charge; $C_\Delta = 0$ iff the record is Δ -lag causally simulatable from S . The identity is exact for every record in $\mathcal{F}_0$ and every Δ (numerical residual $\sim 10^{-14}$); for the exact U -coupled extension and its failure outside the family see the counter-values above.

(ii) **Collapse is a feasibility face, not an impossibility.** Records that collapse ($C_\Delta = 0$) while carrying information exist — e.g. any V -free (N -only) record — so universal non-collapse is false as first conjectured. The correct statement: for $\tau^2 > 0$, $\Delta \leq n - 2$ and

$$D < \rho^2 \frac{n - \Delta - 1}{n},$$

every D -feasible collapsing record must be V -free on all cells $t \leq n - \Delta - 1$, which the distortion budget forbids; at $D \geq \rho^2(n - \Delta - 1)/n$ a feasible collapsing boundary record exists (exhibited at $\Delta = 6$, $n = 16$, $D = 0.3$). Strictness of $\min L_a > \text{block}/n$ in the sub-boundary regime routes through block-optimality plus $C_\Delta > 0$ at the (per-mode unique) block optimizer — not through universal non-collapse.

- (iii) **Class-conditional sandwich, strictly decreasing.** With $\text{block}_n = \min \frac{1}{n} I(T^n; \hat{Y}^n | S^n)$ and $\text{slice}(\Delta)$ the netted single-letter slice coordinate,

$$\text{block}_n < \min_{\text{records}} L_a(\Delta) \leq L_a^{\text{diag}}(\Delta) < \text{slice}(\Delta),$$

where L_a^{diag} is the diagonal-class value; the upper bounds are strictly decreasing in Δ (v0.1 said increasing — a sign error: more prefix access can only cheapen causal erasure), and the lower margin closes geometrically at rate $\lambda_s^{2\Delta}$ (overstatement of the diag-class bound over the true minimum $\leq 1.07 \times 10^{-3}$ at $\Delta = 2$, $\leq 2.4 \times 10^{-6}$ at $\Delta = 5$; $n = 16$, $D = 0.3$).

- (iv) **The smoothing pole — cell-local scope.** For time-local (cell-diagonal) records, the per-interior-cell leakage ratio $\text{leak}_t(\Delta)/\text{leak}_t(\Delta + 1) \rightarrow \lambda_s^{-2}$ in the window interior; the aggregate per-symbol ratio carries the finite-window cell-count prefactor $(n - \Delta - 1)/(n - \Delta - 2)$. The exponent is record-dependent: code-mixing records measurably decay slower (the reverse-water-filling record shows a stable $\approx 0.255/\text{lag}$, empirically $\approx \lambda_s^2/\rho^2$ — characterization open). No constant measured ratio is quoted — the pole is an asymptotic with a named prefactor and a named scope.

- (v) **Not bookkeeping.** L_a is not the fixed-lag static quadratic: the deficit of the bookkeeping candidate is n -pinned, $\text{gap}_n = \text{static}(q_{\text{path}}) - \text{block}_n = 0.0213/0.0263/0.0280$ at $n = 8/16/24$, converging to the spectral gain ≈ 0.0313 ($\text{block}_\infty = 0.529950$ by per-frequency equal-slope allocation; corrected value — see the dated erratum, Remark 10). Record memory and cross-cell code structure carry real value.

Remark 1 (Class scoping — load-bearing). The diagonal (stationary-symmetric) class is *provably suboptimal* for L_a : at $\Delta = 0$, $n = 16$, $D = 0.3$ the first-order optimality certificate fails at the diagonal optimum (feasible projected slope 8.6×10^{-3} against 10^{-6} noise) and explicit non-diagonal records achieve $\min L_a(0) \leq 0.567353 < 0.572255$ (an improvement of 4.9×10^{-3} bits, two independent parameterizations agreeing to 8×10^{-5}); the full-space search behind Conjecture 1 has since sharpened the beat to $\min L_a(0) \leq 0.5667581$ at $n = 16$ (Remark 8). Diagonal optimality *remains proven* for the block coordinate. Every quoted minimum is therefore a class-conditional or finite- n upper bound inside the bracket of (iii); presenting 0.572255 as *the* minimum would be an error.

Remark 2 (Rejected definition). Definition (b) (memoryless eraser) is monotone but has the wrong $\Delta \rightarrow \infty$ invariant (floors 0.0136 bits above the netted block coordinate) and is chain-rule-inconsistent; (c) (Kostina–Hassibi causally-conditioned DI) collapses to the full-path endpoint in Gaussian settings. Definition (a) is the thermodynamically operational coordinate pending a reset-protocol construction (open).

3 The convex program: Theorems R and C

Status: lower-bound prover + independent R-IND-5 verification (own evaluator, own polish, own moment box, fresh seeds; representation identity to 2.0×10^{-12} over 279 cells including edges; the composition chain of Theorem 2 re-proved independently; 11,200 verifier Jensen midpoints and 1,566 curvature probes with zero violations, on top of the prover's 32,000). Harness: `experiments/go14_convexity.py`. Prereg 079 pending — nothing in this section is sealed yet.

Moment coordinates. Write $W = (V^n; Y^n)$ (so $T^n = W$), $\Sigma_W = \text{Cov}(W)$, and for a record $\hat{Y} = AW + Z$ of the family $\mathcal{F}_0$ set

$$H = \text{Cov}(\hat{Y}, W) = A\Sigma_W \quad (n \times 2n), \quad \Gamma = \text{Cov}(\hat{Y}) \quad (n \times n).$$

The family in these coordinates is the convex cone

$$\mathcal{D} = \{(H, \Gamma) : \Gamma - H\Sigma_W^{-1}H^\top \succeq 0\}$$

(the slack is $N_{\text{cov}} = \text{Cov}(Z)$), and the distortion is *affine*: $n \text{ dist} = \text{tr} \Sigma_Y - 2 \text{tr}(H \Sigma_W^{-1} \Sigma_{WY}) + \text{tr} \Gamma$. Write $P = \Sigma_W^{-1}$ and $Q = \Sigma_W^{-1} \Sigma_{WS} \Sigma_S^{-1} \Sigma_{WS}^\top \Sigma_W^{-1}$.

Theorem 1 (R — moment-coordinate representation, schedule-general). Assume Hypothesis 1. On the window $t = 1, \dots, n$ let the eraser of cell t hold $S^{se(t)}$ for an *arbitrary nondecreasing access schedule* $se(\cdot)$, and define the schedule-general counts

$$k(j) = \#\{t : se(t) < j\}, \quad 1 \leq j \leq n$$

(the number of record cells already erased when S_j enters the interleaved order). Then for every $(H, \Gamma) \in \text{int } \mathcal{D}$:

$$\begin{aligned} 2 \ln 2 \cdot n L_a &= \left[\ln \det(\Gamma - HQH^\top) - \ln \det(\Gamma - HPH^\top) \right] \\ &\quad + \sum_{j=1}^n \left[\ln \text{Var}(S_j \mid S^{j-1}) - \ln \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)}) \right]. \end{aligned}$$

Staircase reduction (explicit). For the coordinate's own schedule $se(t) = \min(t + \Delta, n)$ the counts reduce to $k(j) = \max(0, j - \Delta - 1)$. The reduced form is *prefix-staircase-specific*: on non-staircase nondecreasing schedules only the general $k(j) = \#\{t : se(t) < j\}$ is correct — blind use of the staircase formula deviates by up to 1.05 bits (verifier-measured), and off-by-one variants of k are detected at 0.1–0.7 bits.

Proof skeleton. (1) Factor the joint $\ln \det$ of $(S^n, \hat{Y}^n)$ along the interleaved order twice: collecting the $\hat{Y}$ -pivots gives $\sum_t \ln \text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$; collecting the S -pivots gives $\sum_j \ln \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ — this regrouping is the whole trick (and the reason single-cell convexity is never needed). (2) U-independence enters *exactly once* as a premise — the residuals (U, Z) are independent given W — but is load-bearing in *two* legs of the computation: $\text{Cov}(S, \hat{Y}) = KH^\top$ with $K = \Sigma_{WS}^\top \Sigma_W^{-1}$ (numerator/leak leg, giving the Q -Schur complement $\Gamma - HQH^\top$ since $Q = K^\top \Sigma_S^{-1} K$), and $\ln \det \text{Cov}(\hat{Y} \mid W, S^\bullet) = \ln \det(\Gamma - HPH^\top)$ (denominator leg: conditioning on S adds nothing given W). (3) The unconditional S -pivots telescope against $\ln \det \Sigma_S$. This re-proves the 076 chain rule of Theorem 1(i) in one line: the bracket is the block term $I(T^n; \hat{Y}^n \mid S^n)$ and the j -sum is the leakage charge C_Δ in its S -side (reversed) form. Verification: residual 1.15×10^{-12} over 144 prover cells and 2.0×10^{-12} over 279 verifier cells including the edges $\Delta \in \{0, n-2, n-1, n\}$ and random non-staircase schedules. $\square$

Remark 3 (U-independence: one premise, two legs — and the non-conflation rule). Correcting only one leg does *not* restore the identity: at fixed dense- U -coupled records ($n = 12$) the leg-by-leg counter-values are 0.32–2.01 bits (as-stated, numerator-only- corrected, and denominator-only-corrected all break; the harness’s pinned record measures 2.68/3.51/0.83 bits respectively), while correcting *both* legs restores exactness for arbitrary A_u including anticausal couplings (residual at machine precision). Non-conflation: Δ -lag-causal U -coupling ($A_u[t, s] = 0$ for $s > t + \Delta$) extends the *record-space* chain rule of Theorem 1(i) exactly, but does *not* rescue the moment-form Theorem R as stated — measured moment-form residuals 2.4–3.5 bits at Δ -causal couplings (harness pinned record: chain-rule residual 2×10^{-16} , moment-form residual 3.22 bits at $\Delta = 2$). The two extension results must not be conflated.

Theorem 2 (C — joint convexity; KKT points are global; strong duality). On $\mathcal{D}$: (i) the block bracket $B(H, \Gamma) = \ln \det(\Gamma - HQH^\top) - \ln \det(\Gamma - HPH^\top)$ equals $-\ln \det(I - Z)$ with $Z = R^{1/2}H^\top(\Gamma - HQH^\top)^{-1}HR^{1/2}$ and

$$R = P - Q = \Sigma_W^{-1} \Sigma_{W|S} \Sigma_W^{-1} \succeq 0;$$

under the mapping $C \leftrightarrow H^\top$, $V \leftrightarrow \Gamma$ this is *exactly* the 074 convexity lemma’s G -form (GO-13 Theorem 5, block Schur lift), so Z is jointly matrix-convex on $\mathcal{D}$ with $Z \prec I$ on $\text{int } \mathcal{D}$ (Sylvester: $\det(I - Z) = \det M_P / \det M_Q$), and B is jointly convex. (ii) Each leak term $-\ln s_j$ is convex: the pivot $s_j = \text{Var}(S_j \mid S^{j-1}, \hat{Y}^{k(j)})$ is a Schur pivot of a joint covariance that is *affine* in (H, Γ) , hence an infimum of affine functions of (H, Γ) — concave — and bounded below by $\tau^2 > 0$ (the eraser never sees U_j through a U -independent record). (iii) The distortion is affine and $\mathcal{D}$ is a convex cone. Hence nL_a is jointly convex on $\mathcal{D}$; every KKT point of the distortion-constrained program is the *global* family minimum; Slater’s condition holds (interior scalar records), so strong duality makes lower bounds computable: any $\mu \geq 0$ and any point x_p give the certificate $p^* \geq \phi_\mu(x_p) - \|\nabla \phi_\mu(x_p)\| R_{\text{box}}$, with R_{box} from the per-cell moment bounds $\Gamma_{tt} \leq (1 + \sqrt{nD})^2$, $|H_{tj}| \leq \sqrt{\Gamma_{tt}}$ (verifier: the box is valid and nearly tight — an adversarial one-cell record reaches 10.1748 against the box’s 10.1818).

Proof skeleton. (i) is the 074 lift verbatim after the transpose-convention check ($C^\top PC$ with $C = H^\top$ is $HPH^\top$ — mapping exact to 0.0); $R \succeq 0$ by the displayed closed form (verified to 1.7×10^{-14} , $\text{eigmin}(Q) \geq -10^{-15}$, PSD to machine). (ii) is inf-of-affine concavity plus $-\ln$ composition on $[\tau^2, \infty)$. (iii) is inspection. Numerics on record: zero Jensen-midpoint violations (prover 32,000 + verifier 11,200 + 1,566 curvature probes), tangent plane at the (16, 0) winner 400/400 with min slack $+1.13 \times 10^{-2}$, analytic gradient vs FD 1.2×10^{-10} . $\square$

Remark 4 (Per-cell convexity is open — the regrouping is the proof). Theorem 2 proves convexity of the *regrouped* pieces (block bracket, S -side leak sum). Whether the individual per-cell CMI terms of L_a are themselves convex in (H, Γ) remains *open*: no violation was found in any sampling (prover and verifier both, zero hits), but no proof exists either. Nothing in this section relies on per-cell convexity.

Remark 5 (The diagonal class is a convex linear section — its ladder is UB-only because class $<$ family). In (H, Γ) the diagonal (stationary-symmetric) class is a *convex (linear) section* of $\mathcal{D}$: midpoints of diagonal records stay in class (verifier: off-diagonal mass in the C_v eigenbasis at (H, Γ) -midpoints is 0 to machine precision). The earlier “non-convex slice” wording is refuted as worded — what is non-convex is the (g, h, z) *parameter chart* (explicit midpoint witness on record), a different statement. Consequently the diagonal ladder of §4 is upper-bound-only *because the class is a strict subset of the family* (Remark on class scoping, §2), not because of any slice non-convexity;

and within-class values are certifiable by the same Lagrangian machinery restricted to the section. **That upgrade path has since been executed:** all nine ladder cells and three block values are now certified two-sided *class* values (§5.2). Certifying the class does not certify the family — the ladder remains an upper bound on $\min L_a$ because $\text{class} < \text{family}$.

Certificates on record (per- n). $\text{eigmin}(P - Q) = 1.0382 \times 10^{-2} / 5.5944 \times 10^{-3} / 4.4309 \times 10^{-3} / 3.9529 \times 10^{-3}$ at $n = 8/16/24/32$ — *decreasing in n* ; quoted as “ $\geq 3.9 \times 10^{-3}$ for $n \leq 32$ ” (no n -uniform floor is claimed; the earlier single-number quote is an erratum, corrected here). The noise-floor parameterization is inactive at every winner and every polished point.

3.1 Certified two-sided anchors

Lagrangian polish + moment-box tangent bound (Theorem 2), prover (P) and independent verifier (V) each end-to-end; per anchor the tighter certified width of the two is cited. Upper ends are the (feasibility-projected) winner values.

(n, Δ)	certified bracket for p^*	width	source
(16, 0)	[0.566754682, 0.5667581350]	3.45×10^{-6}	V
(16, 1)	[0.5408029, 0.5408064658]	3.6×10^{-6}	P
(16, 2)	[0.5358901, 0.5358939352]	3.8×10^{-6}	P
(24, 0)	[0.5654096, 0.5654138570]	4.3×10^{-6}	P
(24, 1)	[0.5393228, 0.5393377781]	1.5×10^{-5}	P
(24, 2)	[0.5342628, 0.5342788038]	1.6×10^{-5}	P
(32, 0)	[0.5647249, 0.5647418676]	1.7×10^{-5}	P

The 078 anchors are thereby *certified family values*, no longer upper bounds alone; minimizer *uniqueness* is still not claimed (no flat was found — three polishes agree to 2.8×10^{-14} in value and 2×10^{-6} in coordinates — but gauge flats are not excluded; “an optimizer”, never “the”). The harness reproduces four of the brackets end-to-end from cold starts by μ -bisection (convexity makes warm starts a convenience, not a dependence), at widths $4.0/3.0/3.8/9.2 \times 10^{-6}$ for $(16, 0)/(16, 1)/(24, 0)/(32, 0)$ and a block_{16} bracket of width 1.7×10^{-6} .

Corollary 1 (Certified sandwich margin at $n = 16$, $D = 0.3$).

$$\min L_a(0) - \text{block}_{16} \in [0.0317872, 0.0317906]$$

(*verifier-tightened interval*). The block side inherits the sealed block_{16} precision through the proven diagonal optimality of the block coordinate.

Remark 6 (Feasible projection of winner upper ends). The stored winner endpoints are distortion-infeasible by $\sim 4 \times 10^{-11}$ (raw optimizer termination). The one-line feasible projection (blend toward the near-copy record) corrects the upper ends by $\sim 10^{-10}$ — immaterial at the 3.5×10^{-6} width scale; all upper ends above are to be read through this remark.

Remark 7 (“Certified” means floating-point certificates). All brackets here are *floating-point* certificates: the weak-duality and tangent-bound inequalities are exact in real arithmetic and are evaluated in IEEE double precision without interval arithmetic. The certified widths exceed the $f64$ evaluation error by roughly seven orders of magnitude; no claim of formally verified arithmetic is made.

4 The process limit

Everything in this section is under Hypothesis 1 and at the netted parameters, $D = 0.3$. Status: probe + independent R-IND-5 verification (two evaluator routes agreeing to 2×10^{-10} ; all sealed 076 anchors reproduced); pre-registration (078) pending — no numerical face here is sealed yet.

Conjecture 1 (Process limit of the causal-erasure coordinate — v0.2, R-IND-5-scoped). *Let $L_n^{\text{fs}}(\Delta) = \min L_a(\Delta)$ over the full (non-diagonal) family $\mathcal{F}_0$ on the window of length n , and $L_n^{\text{diag}}(\Delta)$ the diagonal-class value.*

- (1) **Existence at $O(1/n)$.** *Both coordinates converge as $n \rightarrow \infty$ with $O(1/n)$ finite-window drift (model attack: fitted exponent $\alpha = 0.996\text{--}1.001$). Two-point $1/n$ limits, calibrated on the block face ($\leq 5 \times 10^{-6}$): full space $L_\infty^{\text{fs}}(\Delta) = 0.562725/0.536400/0.531049$ at $\Delta = 0/1/2$ (three- n consistency at $\Delta = 0$: $0.5627253/0.5627259/0.5627265$ from $n = 16/24/32$ pairs); diagonal class $0.568571/0.537401/0.531206/0.529977$ at $\Delta = 0/1/2/4$ ($\pm \leq 1.2 \times 10^{-5}$).*
- (2) **Closure rate, constant unclaimed.** *The excess $E_\infty(\Delta) = L_\infty^{\text{diag}}(\Delta) - \text{block}_\infty$ obeys $E_\infty(\Delta) \sim c \lambda_s^{2\Delta}$ as a $\Delta \rightarrow \infty$ asymptotic only: the per-lag ratio rises toward $\lambda_s^{-2} = 7.955$ from below ($5.18 \rightarrow 5.93 \rightarrow 6.87 \rightarrow 7.65 \rightarrow$ within error at $\Delta = 5 \rightarrow 6$) and is unresolved beyond $\Delta \approx 6$; the constant is quoted as the bracket $c_{\text{diag}} \in [0.105, 0.125]$ only — $c(\Delta) = 0.107/0.112/0.117$ at $\Delta = 4/5/6$, still rising, not converged through $\Delta = 6$. The full-family constant is data-supported only as $c_{\text{fs}} \in (0.07, 0.125]$. No single-letter identification of c is claimed.*
- (3) **Strictly above every tested causal-conditioning spectrum — certified at every computed n ; in the limit conditional on (H^*) at $\Delta = 0$ and on the $O(1/n)$ model at $\Delta = 1, 2$.** *At every computed anchor ($n \in \{16, 24, 32\}$, $\Delta \in \{0, 1, 2\}$) the certified lower bracket end of §3.1 sits strictly above the causal-spectral allocation value (Remark 9) — a certificate-strict gap (smallest exceedance $\approx 4 \times 10^{-3}$ bits at $(24, 2)$, $\geq 250\times$ the bracket width there), not an optimizer report. In the limit, the plateaus $+0.014781/+0.004057/+0.000795$ ($\Delta = 0/1/2$) above the allocation are $160\text{--}3000\times$ the calibrated extrapolation error and carry the wrong shape (drifting per-lag rate, not the allocation's clean λ -law). At $\Delta = 1, 2$ the limit statement remains conditional on the $O(1/n)$ finite-window model. At $\Delta = 0$ it is no longer model-conditional: Corollary 3 derives $L^\infty(0) \geq 0.5515989 > 0.5479448$ from the certified $n = 32$ bracket and the transfer constant, conditional instead on Hypothesis 2 (H^*). The sub-additivity half of the old n -monotonicity face is now closed unconditionally (Theorem 3); what remains open is the boundary charge at optimizers (Question 1). Both closed-form candidates ($c_{\text{spec}} = 0.0190 \pm 0.0007$, $c_{\text{static}} = 0.026668$) are too small at every Δ . No claim is made that no single-letter form exists. The gap is genuine cross-cell code value: the winner cuts leakage $0.033 \rightarrow 0.024$ by paying block penalty the per-frequency relaxations cannot express.*
- (4) **Mechanism (testable).** *At every full-space optimizer computed ($n = 16/24/32$, $\Delta = 0/1/2$; seven adversarial starts converge to the same record, KKT certificate 8.4×10^{-6}): the Y -coupling and noise kernels are time-symmetric, where time-symmetric means transpose symmetry $A_y = A_y^\top$ (relative residual $< 5 \times 10^{-5}$; the stronger persymmetry $JA_y^\top J = A_y$ fails at the $\approx 6\%$ level at the window edges — transpose symmetry is the invariant); all causal structure lives in the V -coupling A_v , whose interior-row kernel changes sign exactly at lag Δ*

— *horizon-matched V-cancellation: the record aligns V-content with the span the eraser will hold and anti-correlates V beyond the access horizon, suppressing smoothing leakage. This is exactly what the symmetric-by-class diagonal family cannot express (and explains the 076 beat). Falsifiable per-cell at any n .*

Remark 8 (Reproducible anchors — raw finite- n values only). The anchors of record are *raw finite- n L_a values*: family min $L_a(0) \leq 0.5667581$ at $n = 16$ (beats the 076-recorded 0.567353 by 5.9×10^{-4} ; all start basins converge to it within 10^{-7}); full-space $L_a(0) \leq 0.56474187$ at $n = 32$; the diagonal ladder value $L_a^{\text{diag}}(n=48, \Delta=6) = 0.5316222730$. The full-space upper ends are now *certified two-sided* (§3.1); the diagonal-ladder values are now certified two-sided *within the class* (§5.2) and remain class-conditional upper bounds on the family minimum (Remark 5). The extrapolated $E_\infty(6)$ is quoted *only* with an explicit error bar — $E_\infty(6) \approx 4.4 \times 10^{-7} \pm 10\%$ (two-point pair sensitivity); it is *not* a four-digit anchor, the raw $L_a(48, 6)$ above is.

Remark 9 (The causal-spectral allocation is a reference construction). The causal-spectral allocation: equal-slope per-frequency conditional quadratics with conditioning spectrum the Δ -lag causal Wiener error spectrum $S_e^{(\Delta)}$. Values 0.547945/0.532344/0.530253 at $\Delta = 0/1/2$, verified to seven digits by an independent (time-domain) route; it reproduces block_∞ at the two-sided endpoint to 1.6×10^{-10} , and its own excess obeys the $\lambda_s^{2\Delta}$ law cleanly. It is a *reference construction only*: it carries no achievability map (not an upper bound on min L_a) and no data-processing direction (not a lower bound). Item (3) of Conjecture 1 is a statement about mismatch with the tested spectra, never a bound.

Remark 10 (Erratum — dated 2026-08-05). v0.2 of this document quoted $\text{block}_\infty = 0.52991$ in Theorem 1(v), propagated from the pass-1 coarse frequency waterfilling; the converged per-frequency equal-slope value is $\text{block}_\infty = 0.529950$ (corrected in item (v) above; the spectral gain ≈ 0.0313 is unaffected). Same date, probe-record erratum: the process-limit probe’s diagonal excess at $\Delta = 4$ is 2.66×10^{-5} , not 2.75×10^{-5} .

Dated 2026-08-06 (transfer note). Three further corrections, all carried into §5: (a) the constant was recorded as $c(1) \leq 1.1257$; the computed value is 1.1257294, so the recorded bound is false in the fifth decimal and $c(1) \leq 1.1258$ is used throughout (Remark 12). (b) The transfer note cites a value 0.564729705 for the certified lower end at $(32, 0)$; no computation on record produces it. The sealed value is $\text{LB}(32, 0) = 0.5647327869$ (`results/G014-convexity.json`, `s5_32_0`), and it is the only one used in Corollary 3; 0.564729705 must not be cited. (c) The plateau margin is $215\times$ the quoted bracket width, not the $\approx 250\times$ of the transfer note.

5 The n -transfer lemma: Theorem T and the hypothesis (H^*)

Status: prover + independent R-IND-5 verification (own `slogdet` evaluator, own certified anchors, own re-derivation of the constants and of the corollary arithmetic; six mandatory restatements, all in force in this revision). Harness `experiments/go14-transfer.py`. Registration 080 pending — nothing in this section is sealed.

Write $\phi_n(\Delta) = \inf_{\mathcal{F}_0} L_a(\Delta)$ for the family value on the window of length n (certified two-sided in §3.1) and $f(n) = n \phi_n(\Delta)$. Let

$$\kappa = \frac{1}{2} \log_2 \frac{1}{1-a^2} = 0.736966 \text{ bits}$$

be the AR(1) boundary information at the netted pole $a = 0.8$. For a split of the n -window into blocks b_1, b_2 define the *per-side boundary charges*

$$D_i = \sum_{t \in b_i} \left[I(T^{b_i}; \hat{Y}_t \mid \text{own}C_t) - I(T^{b_i}; \hat{Y}_t \mid \text{big}C_t) \right] \geq 0,$$

where $\text{own}C_t$ is the conditioning set of cell t inside its own block and $\text{big}C_t = S^{se(t)}, \hat{Y}^{t-1}$ is the big-window one; D_i is exactly what the split loses at the boundary. Write $E = (S^{b_1}, \hat{Y}^{b_1})$ for the block-1 record-and-reference history that the block-2 cells see in $\text{big}C$ but not in $\text{own}C$.

Theorem 3 (T — exact subadditivity, Fekete existence, and the lower transfer side; *unconditional*). Fix Δ and assume Hypothesis 1.

- (i) **Exact subadditivity.** $f(n_1 + n_2) \leq f(n_1) + f(n_2)$ for all $n_1, n_2 \geq 1$. Take ε -optimal records of the two windows, independent of each other, and concatenate: the result is a feasible record of the $(n_1 + n_2)$ -window (the distortion constraint is the cell average), and *at every cell* the big-window per-cell term is at most the own-window term. The numerator $\text{Var}(\hat{Y}_t \mid S^{se(t)}, \hat{Y}^{t-1})$ can only shrink when the conditioning set grows from the own block to the whole window; the denominator $\text{Var}(\hat{Y}_t \mid T^n, S^{se(t)}, \hat{Y}^{t-1})$ is *unchanged*, because a block-diagonal record makes cell t conditionally independent of the other block given its own block's T — verified as an exact equality (relative deviation 2×10^{-14} over all cells, including the unequal split $8 + 4 \rightarrow 12$). Letting $\varepsilon \downarrow 0$ gives the inequality for the infima.

- (ii) **Fekete.** $\phi_n \geq 0$, because each per-cell term is a conditional mutual information; hence $f \geq 0$ is subadditive and

$$L^\infty(\Delta) := \lim_{n \rightarrow \infty} \phi_n(\Delta) = \inf_n \phi_n(\Delta)$$

exists in $[0, \infty)$.

Hygiene (both steps were demanded by R-IND-5 and are stated, not assumed). (a) The non-negativity $\phi_n \geq 0$ is what makes Fekete's lemma give a finite limit rather than $-\infty$; it is immediate from the CMI form of the coordinate, but it is a hypothesis of the lemma and is recorded as such. (b) The concatenation argument is run on ε -optimal records and the limit $\varepsilon \downarrow 0$ is taken at the end, so the theorem never needs ϕ_n to be *attained*. (Attainment does hold, by Theorems 1 and 2 plus the moment box, but it is not used.)

Corollary 2 (The lower transfer side is a theorem). $0 \leq \phi_n(\Delta) - L^\infty(\Delta)$ for every n — *unconditionally*. Every certified finite- n family value of §3.1 is therefore an upper bound on the limit; in particular $L^\infty(0) \leq \phi_{32}(0) \leq 0.5647419677$.

5.1 The reverse direction is hypothetical

The matching *upper* transfer bound $\phi_n - L^\infty \leq c/n$ needs the boundary charge to be controlled at the optimizers of the doubling chain. It is stated as a numbered hard hypothesis, and *not* as a lemma, for the reason given immediately after it.

Hypothesis 2 ((H*) — boundary charge at dyadic-chain optimizers; hard, load-bearing). *Fix Δ and a base window length n . At every optimizer of the dyadic chain $n \cdot 2^k$, $k = 0, 1, 2, \dots$, the two per-side boundary charges of the split into halves satisfy*

$$D_1 + D_2 \leq c(\Delta; n), \quad c(\Delta; n) = (2 - \mathbf{1}\{\Delta = 0\}) \kappa - I_n, \quad I_n = I(S^{b_1}; S'^{1.. \Delta+1}).$$

Remark 11 (Why (H^*) is a hypothesis: “ $\leq \kappa$ per side” is *refuted* as an $\mathcal{F}_0$ lemma). The natural lemma — that each side’s boundary charge is at most the AR(1) boundary information κ , uniformly over the family — is *false*, and its sharpened form $D_i \leq \kappa - I_n$ is false too. Explicit D -feasible records in $\mathcal{F}_0$ (no U -coupling whatsoever; the whole budget respected) spend block-1 rows on copies of block-2 cells:

- two V -copy rows give $D_2 = 0.4563 > 0.4203 = \kappa - I_8$: the *sharpened* constant is exceeded by a mild, feasible perturbation;
- three Y -copy rows give $D_2 = 18.93 \text{ bits} \approx 26\kappa$ at distortion 0.2986: the charge is not bounded by κ at all.

Nothing stops the construction from being repeated: the worst case over the family is $\Theta(m)$ in the block length. **No universal constant exists, and no proof of the reverse direction can avoid using optimizer structure.** *Therefore: this document never prints “ $\leq \kappa$ per side” as an $\mathcal{F}_0$ lemma.*

The two witnesses above are the verifier’s; they are not in the repository, so the harness pins its own analytic records of the same class and measures $D_2 = 0.7749$ for two V -copy rows (distortion 0.2706) and $D_2 = 13.94 \text{ bits} \approx 19\kappa$ for three Y -copy rows (distortion 0.2754). Same refutation, independent witnesses; the refutation itself is a gated fact of the harness, not a remark.

What *is* measured is that the bound holds comfortably *at optimizers*. At the $\Delta = 0$ family optimizers the split charges are $D_2 = 0.076971$ at $m = 8$ and $D_2 = 0.076926$ at $m = 16$ ($D_1 = 0$ at $\Delta = 0$, the block-1 extension set being empty), i.e. a factor 5.5 inside $c(0) = 0.4202858$; and the boundary information itself is $I(E; T^{b_2}) = 0.543057$, some 0.19 bits under κ , stable between the two m . **These verifications were carried out at $m \in \{8, 16\}$ only;** (H^*) asserts the behaviour along the whole chain, and that assertion is not verified.

Theorem 4 (T^* — boundary-charged reverse direction and the two-sided bound; *conditional on* (H^*)). Fix Δ and a base n , and assume Hypothesis 2 along the chain $n \cdot 2^k$. Then

$$(ii) \quad \phi_m(\Delta) - \phi_{2m}(\Delta) \leq \frac{c(\Delta; n)}{2m} \text{ for every } m = n \cdot 2^k;$$

$$(iii) \quad \text{summing the geometric telescope, } 0 \leq \phi_n(\Delta) - L^\infty(\Delta) \leq \frac{c(\Delta; n)}{n}, \text{ the lower half being Corollary 2 and unconditional.}$$

Exact superadditivity is refuted, so (ii) cannot be sharpened to $c = 0$: the certified anchors are strictly decreasing, $\phi_8 > \phi_{12} > \phi_{16} > \phi_{24} > \phi_{32}$, with the new certified transfer anchors $\phi_8(0) \in [0.570790938, 0.5707933842]$ and $\phi_{12}(0) \in [0.568030932, 0.5681028134]$. The bound is not tight either: the measured drift $n(\phi_n - L^\infty) \approx 0.0645$ at $n = 32$ sits $6.5\times$ inside $c(0) = 0.4203$.

Remark 12 (The constants, and their n -uniformity). $I_n = I(S^{b_1}; S'^{1.. \Delta+1})$ is *monotone increasing in n* , so $c(\Delta; n) = (2 - \mathbf{1}\{\Delta = 0\})\kappa - I_n$ is the supremum of the per-step constants along the whole dyadic chain issuing from base n — which is precisely what the telescope in Theorem 4(iii) requires, and why the constant must be built from I_n and not from I_∞ . Numerically I_n is converged to I_{32} by $n = 16$ (to $< 10^{-12}$), and

$$I_{32} = 0.3166798 / 0.3482018 / 0.3520686 \quad \text{at } \Delta = 0/1/2,$$

each equal to I_∞ to below 10^{-15} . Hence

$$c(0) \leq 0.42029, \quad \boxed{c(1) \leq 1.1258}, \quad c(2) \leq 1.1219.$$

Erratum: the transfer note recorded $c(1) \leq 1.1257$; the computed value is 1.1257294, so 1.1257 is false in the fifth decimal. Use 1.1258.

Remark 13 (The zero-claim and the marginalization step are $\mathcal{F}_0$ -only). Two steps carry the whole cancellation. (a) The *zero-claim* $I(E; S'_j \mid \text{pfx}_j, T^{b_2}) = 0$ makes the interleaved chain rule cancel the record–reference coupling exactly (residual $\leq 2 \times 10^{-14}$ over pinned $\mathcal{F}_0$ records). (b) The *marginalization* step replaces the block-2 sub-record by an effective 8-window record with the same law (deviation 2×10^{-16} ; the effective noise is independent of both W^{b_2} and U^{b_2}). **Both are tagged $\mathcal{F}_0$ -only.** They fail outside the family, and the failure is not a technicality: with block-1 rows coupled to U^{b_2} the zero-claim is violated by 0.2223 bits, and with the coupling confined to *block-2 rows only* it is still violated by 0.0950 bits — in the latter case the violation is induced purely through the conditioning, with no block-1 record touching U at all. So U -independence is load-bearing here in the same way it is load-bearing in Theorem 1 (Remark 3), and for a third time.

Corollary 3 (The $\Delta = 0$ plateau over the causal-spectral allocation — *conditional on (H^*)*). Assume Hypothesis 2 at $\Delta = 0$ with base $n = 32$. With the sealed 079 certified lower end $\text{LB}(32, 0) = 0.5647327869$ (*harness go14_convexity.py, entry s5_32_0*) and $c(0) = 0.4202858$,

$$L^\infty(0) \geq 0.5647327869 - \frac{0.4202858}{32} = 0.5515989 > 0.5479448,$$

the right-hand side being the causal-spectral allocation at $\Delta = 0$ (Remark 9). The margin is $+3.654 \times 10^{-3}$: $215 \times$ the 1.7×10^{-5} bracket width quoted for $(32, 0)$ in the anchor table of §3.1, and $398 \times$ the 9.18×10^{-6} width the sealed harness actually achieved. This is a statement about the limit itself, not about the $O(1/n)$ finite-window model.

The margin is not slack in the constant, and the base matters. *Running the same arithmetic from base $n = 24$ gives $0.5654101216 - 0.4202858/24 = 0.5478982 < 0.5479448$: it fails by 4.7×10^{-5} . (H^*) is genuinely load-bearing, and so is the choice of base window. $\Delta = 1$ and $\Delta = 2$ remain $O(1/n)$ -model-conditional: their constants are $2.7 \times$ larger and their plateaus $3.6/18 \times$ smaller, so a certified anchor at $n \approx 260$ resp. $n \approx 1320$, or a sharper constant, would be needed.*

Remark 14 (The repair path that would remove (H^*)). What is missing is an *optimizer-regularity* bound: any argument that bounds $I(E; T^{b_2})$ at the optimizers of the doubling chain — for instance from the convex-program structure of Theorems 1 and 2, whose KKT system the optimizers satisfy globally — would restore the unconditional label on Theorem 4 and Corollary 3 *with the same constants*. The measured evidence for such a bound is $I(E; T^{b_2}) = 0.543$, stable in m and 0.19 bits under κ . This is the natural next prover face; it is not a numerical gap but a missing theorem.

5.2 The within-class ladder, certified (the Remark 5 upgrade, executed)

The diagonal class is a convex *linear* section of $\mathcal{D}$ (Remark 5), so the Lagrangian machinery of Theorem 2 restricted to the section — with a section moment box $R_{\text{box}} = 73/95/136$ at $n = 24/32/48$, verified attained and hence tight — certifies the ladder two-sided. All nine ladder cells and the three block values are now certified *class* values (per cell the tighter of the two independent end-to-end computations):

(n, Δ)	certified within-class bracket	width
(24, 4)	[0.5333160113, 0.5333170450]	1.0×10^{-6}
(24, 5)	[0.5332969026, 0.5332971560]	2.5×10^{-7}
(24, 6)	[0.5332945088, 0.5332947270]	2.2×10^{-7}
(32, 4)	[0.5324808853, 0.5324814627]	5.8×10^{-7}
(32, 5)	[0.5324606031, 0.5324610143]	4.1×10^{-7}
(32, 6)	[0.5324578912, 0.5324584264]	5.4×10^{-7}
(48, 4)	[0.5316458819, 0.5316462767]	3.9×10^{-7}
(48, 5)	[0.5316244073, 0.5316250208]	6.1×10^{-7}
(48, 6)	[0.5316217661, 0.5316233575]	1.6×10^{-6}
block ₂₄	[0.5332936495, 0.5332949450]	1.3×10^{-6}
block ₃₂	[0.5324577110, 0.5324581027]	3.9×10^{-7}
block ₄₈	[0.5316214213, 0.5316220475]	6.3×10^{-7}

The (48, 6) cell is the loosest (its feasible-projection correction is 1.1×10^{-6} , still inside the bracket), which is the same cell the raw-anchor demotion of Remark 8 concerns.

What this does *not* certify. The excess $E_\infty(\Delta)$ is a *two-point* $1/n$ *extrapolation* of these class values against the block values, so the re-assembled E_∞ intervals are $1/n$ -*model-conditional*, *not certificates*, and are tagged as such: $E_\infty(4) \in [2.50, 2.75] \times 10^{-5}$, confirming the 2.67×10^{-5} of Conjecture 1 in bar; $E_\infty(5) \in [1.7, 4.8] \times 10^{-6}$, sign resolved positive by the tighter re-assembly and *unresolved* by the looser one; and $E_\infty(6) \in [-3.2, 1.5] \times 10^{-6}$, whose bracket *contains zero* at every achievable width. The last of these independently re-ratifies the R-IND-5 demotion of $E_\infty(6)$ to the raw anchor $L_a^{\text{diag}}(48, 6) = 0.5316222730$ (Remark 8). No claim is made here about how many optimizer endpoints were true class minima; the transfer note’s “4/9 endpoints” remark is *withdrawn* as unverifiable from the record.

Question 1 (Open). (1) *The boundary charge at optimizers — what is left of the old n -monotonicity face after Theorem 3 closed its subadditivity half unconditionally. An optimizer-regularity bound on $I(E; T^{b_2})$ along the doubling chain (Remark 14) would discharge Hypothesis 2 and make Theorem 4 and Corollary 3 unconditional with the same constants; a uniform family bound cannot exist (Remark 11). Separately, certified anchors at $n \approx 260/1320$ would carry the $\Delta = 1, 2$ plateaus without (H^*) ;* (2) *per-cell convexity (Remark 4) — the within-class certification of the diagonal ladder (Remark 5) is no longer open, it is executed in §5.2;* (3) *the U -coupled coordinate (Hypothesis 1 dropped): a different, far cheaper object, definitionally open;* (4) *the reset protocol realizing definition (a);* (5) *identification of the closure constant beyond $\Delta \approx 6$, and the innovations characterization of the limit (vector NRDF closed forms were refuted in general — Stavrou–Tanaka–Tatikonda — so only scalar forms are expected cheaply).*

House rules: the novelty gate is *cleared* for the four objects swept (arXiv + DBLP + Cross-ref; S2 residual is belt-and-suspenders only), conditional on the five must-cite adjacents (Berta et al.; Boyd–Mandal–Crutchfield; Venkat–Weissman–Carmon–Shamai; Sandberg et al.; Anderson 2018) — headline novelty language may be promoted only with the five citations in force, and nothing above claims it yet. R-IND-5 passes 1 (+closure), the Conjecture-v0.2 pass, the Theorems-R+C pass (seven mandatory restatements) and the n -transfer pass (six mandatory restatements — the (H^*) demotion, the κ -per-side refutation, the constants, the $\mathcal{F}_0$ -only tags, the corollary arithmetic and the ladder tagging — all in force in this revision) are on record

(go14-causal-erasure-PROBE.md); seal GO-P-2026-076 governs the theorem’s numerical faces; the process-limit faces of §4 await prereg 078 (harness `experiments/go14_process_limit.py`); the convex-program faces of §3 await prereg 079 (harness `experiments/go14_convexity.py`: representation identity incl. non-staircase schedules, PSD/closed-form/Sylvester certificates, 4,200-pair Jensen sweep, tangent plane, cold-start certified brackets at $(16,0)/(16,1)/(24,0)/(32,0)$ + block_{16} , sandwich-margin window, and the two-leg / non-conflation U -gates); the transfer faces of §5 await prereg 080 (harness `experiments/go14_transfer.py`: per-cell subadditivity with block-local denominators, the interleaved-prefix set identity, the $\mathcal{F}_0$ -only zero-claim with its U -coupled counter-values, the (H^*) refutation netted as a gated fact, the optimizer charges at $m \in \{8, 16\}$, the constants and the corollary arithmetic including the failing $n = 24$ base, and reproduction of the transfer anchors and of the within-class ladder). The scope proposed for 080 is: Theorem 3 and Corollary 2 (unconditional), the transfer anchors, the within-class ladder, and the conditional Theorem 4 / Corollary 3 under (H^*) .